Research Work Summary

Semester: Fall, 2019

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PCELab
Predictive Computational Engineering

Outline

1. Additive Manufacturing Model
2. Finite Element Method Automation: FEniCS
3. Phase Field Models
4. High Performance Computing
5. Uncertainty Quantification

Additive Manufacturing Model

1. Acquired general background in direct laser deposition (DLD) for additive manufacturing of metals and alloys
2. Identified computational needs and challenges in powder-bed fusion (PBD, selective laser melting, SLM) and the importance in microstructural evolution
3. Conceptualized a continuum physics based finite element framework for prototyping the forward model

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